Dear Editor,

Please consider this research article, **"Proxy-Based Urban Change Prediction under Spatial Generalization: Improving Tail Reliability with Lightweight Calibration,"** for publication in the *Journal of Applied Remote Sensing.*

This study addresses a practical gap in remote-sensing-enabled urban monitoring: the mismatch between reported model performance and real-world reliability when predictions are transferred to unseen spatial units. Using the CivitasGrid-TLP dataset (2015–2024), we evaluate proxy-based urban change prediction from WorldPop and VIIRS under a leakage-aware GroupSplit(grid_id) protocol and benchmark multiple tabular models together with two tail-oriented strategies: a routing-based Mixture-of-Experts (MoE) and a lightweight post-hoc calibration method.

Our main findings are as follows. First, ΔWorldPop is substantially more predictable than ΔVIIRS, whose year-to-year differencing amplifies observational noise and weakens signal under spatial generalization. Second, tail errors dominate practical model failure, even when average metrics appear acceptable. Third, an RMSE-anchored contrastive affine calibrator consistently improves tail reliability with minimal impact on overall performance across diverse baselines. In contrast, a two-expert MoE does not improve performance and becomes less stable under spatial shift.

I believe the manuscript is well aligned with the *Journal of Applied Remote Sensing* because it focuses on a concrete Earth-observation application, uses widely adopted remote-sensing proxy products, and emphasizes geographically realistic evaluation rather than only in-sample accuracy. The paper offers a reproducible benchmark, highlights an important evaluation pitfall caused by spatial leakage, and proposes a simple, deployable calibration strategy for improving reliability in urban change mapping.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests and no external funding.

Thank you for your consideration.

Sincerely,

Winston Huang